# Shayan Poigai

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## EDUCATION

**Santa Clara University — B.S. in Computer Science** Expected December 2027

*Major GPA: 4.0 / 4.0*

**Relevant Coursework:** Advanced Algorithms, Data Structures & Algorithms, Advanced C++, Theory of Automata, Linear Algebra, Discrete Math

## WORK EXPERIENCE

**Co-Founder & Software Engineer** May 2026 – Present

*Adorus (adorusjewels.com)*

- Sole developer on a 4-person founding team, building the full-stack jewelry e-commerce platform end-to-end (pre-launch) while co-founders lead sales and marketing.
- Shipping features rapidly using **AI-assisted development workflows (Claude Code)** to accelerate coding, refactoring, and debugging while reviewing and validating all generated output.
- Translating co-founder stakeholder feedback into data models, API contracts, and component architecture for the platform
- Deploying a responsive **React.js/Node.js** frontend backed by **Java Spring Boot** RESTful services and a **PostgreSQL** database, deployed on **AWS EC2**, with **AWS S3** implemented for secure, cloud-based product image storage.

**Incoming Software Engineering Intern** Starts June 2026

*HerbsPro*

- Joining the platform team to build backend features across the e-commerce stack.

**Machine Learning Research Assistant** Dec 2025 – June 2026

*Santa Clara University*

- Built an **NLP/data pipeline** analyzing 5,000+ scraped posts from Twitter, Reddit, and YouTube to study public sentiment toward LLMs as a therapeutic tool, applying **BERTopic, LDA, and NMF topic modeling.**
- Engineered **Pandas** preprocessing to normalize unstructured text, reducing noise in high-dimensional social media datasets.
- Produced **Matplotlib / Seaborn** analytics on shifting attitudes toward AI mental-health support; co-authored a paper as primary writer, submitted to **IEEE/ACM ASONAM 2026** (results expected July 2026).

**CS Teaching Assistant & Algorithms Grader** Sept 2025 – June 2026

*Santa Clara University*

- Mentored 40+ students in **C++** (memory management, pointers, OOP) and debugged logic errors live using **GDB**
- Graded Theory of Algorithms coursework for 40+ students across 2 sections, evaluating runtime analysis, greedy, divide-and-conquer, and dynamic-programming solutions.

## PROJECTS

**Disease Tracker** Oct 2025

*React, FastAPI, AWS Bedrock, DynamoDB* Github

- Built a **FastAPI REST API** serving real-time disease metrics and risk assessments with low-latency responses.
- Engineered an **AWS Bedrock** AI risk-scoring pipeline and designed **DynamoDB** schemas for scalable data ingestion.

**Kuhn Poker vs Quantum** June 2026

*Qiskit, IBM Quantum, FastAPI, React* Github, Live

- Built a full-stack game where a 12 parameter, 6-qubit variational quantum circuit (superposition + entanglement) derives the opponent's mixed strategy, deployed on **Vercel** and **Render**.
- Ran the circuit on real **IBM quantum hardware** (e.g., ibm_fez); real-chip probabilities matched simulation within ~1.7 percentage points on average on a 4,096-shot run.

**Social Network (C++ / Qt)** Sept 2025

*C++, Qt, MVC, Graphs* Github

- Built a desktop social app with a **Qt GUI** and **MVC** architecture; modeled friendships with a custom graph and generated friend suggestions via **BFS** traversal, with file **I/O** for persistence.

**Agentic RAG Pipeline** July 2025

*Python, ChromaDB, Ollama, MCP* Github

- Built a modular **RAG** framework over **ChromaDB** and **Ollama** for semantic search across dynamic datasets.
- Implemented an **MCP** (Model Context Protocol) server exposing local data as tools for AI agents via JSON-RPC, designed for drop-in integration of new datasets.

## TECHNICAL SKILLS

**Languages:** C++, Python, Java, JavaScript, SQL

**Frameworks & Tools:** React.js, Node.js, Spring Boot, FastAPI, Docker, Linux, Qt, Git, Vercel

**Cloud & Data:** AWS (EC2, S3, Bedrock), PostgreSQL, DynamoDB, ChromaDB

**AI / ML & Dev Tools:** RAG, MCP, NLP, Ollama, Pandas, Qiskit, topic modeling (BERTopic, LDA, NMF); Claude Code, ChatGPT